Political Communication and Market Volatility

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Introduction & Contribution

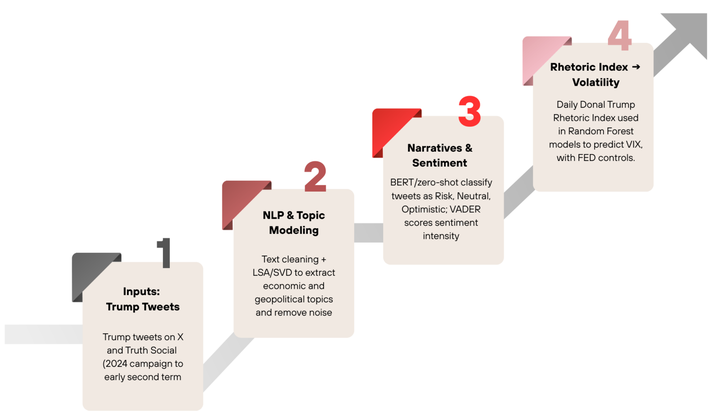
Financial markets are highly sensitive to political information that alters expectations about the future government decisions and macroeconomic conditions (Pástor & Veronesi, 2012, 2013). The asset-pricing literature shows that political uncertainty affects expected returns and risk premia, and that instability in economic policy is systematically associated with higher volatility (Baker et al., 2016). Empirical evidence also indicates that electoral cycles and the broader institutional environment depress corporate investment, revealing that real activity responds to these shocks (Julio & Yook, 2012). The text-as-data literature establishes that unstructured textual information contains predictive signals for macroeconomic indicators (Tetlock, 2007; Kelly et al., 2021). Recent studies further suggest that presidential tweets affect stock market volatility (Nishimura et al., 2021; Kinyua et al., 2021). However, these effects are measured at *intra-day* frequency. Much less is known about whether tweet-based presidential rhetoric, characterized by sudden and informal disclosures, generates next-day close-to-close predictability.

In this framework, we ask **whether sudden and unexpected disclosures in Donald Trump's tweets generate high-frequency uncertainty shocks that predict daily VIX changes.** This project contributes by bringing high-frequency digital presidential communication into the literature on political uncertainty and financial markets. In doing so, it integrates modern NLP tools with financial econometrics to construct a daily Presidential Rhetoric Index and evaluates its predictive content for implied volatility using OLS with HAC standard errors and an event study design based on rolling Z-scores.

Data

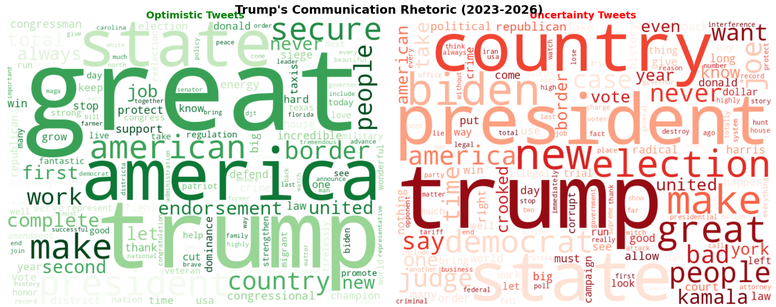
Our dataset comprises 16,492 tweets from Donald Trump on X and Truth Social, spanning from November 5, 2023, to March 16, 2026. Each tweet includes the date, time, URL, and original text. We employed a robust preprocessing pipeline, including lowercasing, removal of URLs, mentions, hashtags, and punctuation, followed by lemmatization and stopword removal using spaCy. This resulted in 11,863 clean, non-empty tweets for analysis. To measure market volatility, we downloaded daily VIX data from Yahoo Finance, covering the same period. This VIX data was carefully matched with the tweets, accounting for market closing times, ensuring tweets published after 4 PM ET were matched with the next day's VIX, yielding a balanced daily panel of **848 trading-day observations**.

Methodology



Sentiment Analysis

We utilized two distinct sentiment analysis approaches: VADER (Valence Aware Dictionary and Sentiment Reasoner) for its ability to analyze social media text, providing 'compound', 'positive', 'negative', and 'neutral' scores for each tweet. Additionally, we implemented Zero-Shot BERT classification using 'facebook/bart-large-mnli' to categorize tweets into 'Optimistic', 'Neutral', or 'Risk' categories, based on their potential impact on financial markets. The word clouds below show the dominant vocabulary in *Optimistic* tweets (left) and *Risk* tweets (right), illustrating the thematic contrast between rhetoric types. Optimistic tweets concentrate around economic achievement and patriotic language, while risk tweets cluster around conflict, threats, and political confrontation.



OLS regression of ΔVIX on presidential rhetoric

$$\Delta\text{VIX}_t = \alpha + \beta_1\,\text{RhetIdx}_t + \beta_2\,n\text{Tweets}_t + \beta_3\,\text{Fed}_t + \beta_4\,\Delta\text{VIX}_{t-1} + \varepsilon_t$$

Main Results

Our OLS analysis reveals that the Rhetoric Index and tweet volume are not statistically significant predictors of next-day VIX changes. The only robust predictor is lagged volatility ($\beta = -0.061$, $p = 0.023$), reflecting mean-reversion in implied volatility.

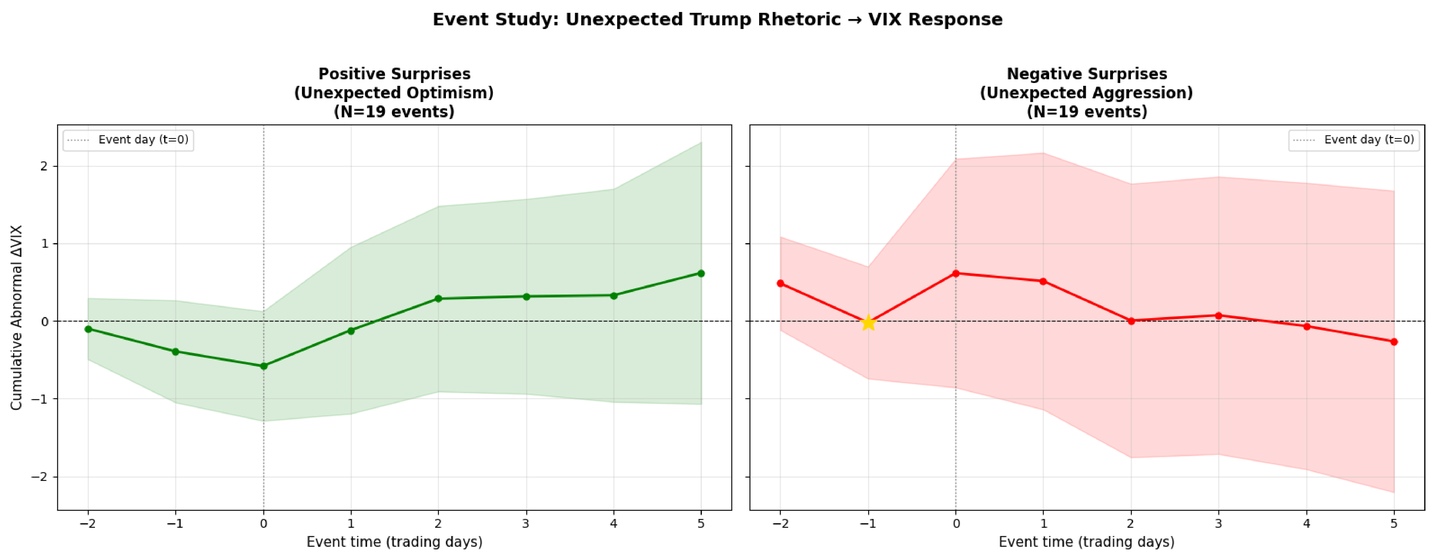
	Coef.	Std. Err.	z	p-value
Intercept	1.044**	0.464	2.252	0.024
Rhetoric Index	-0.005	0.137	-0.038	0.969
N. Tweets	-0.001	0.004	-0.163	0.871
Fed Communication	0.042	0.130	0.319	0.749
ΔVIX_{t-1}	-0.061**	0.027	-2.270	0.023
N	848			
R^2	0.030			

**$p < 0.05$. HAC SE, 5 lags.

As a robustness check, replacing the combined index with decomposed measures (VADER and uncertainty ratio separately) raises R^2 marginally to 0.041; *vader_mean* ($p = 0.075$) and uncertainty ratio ($p = 0.107$) are borderline significant at 10%, but neither reaches the 5% threshold. The overall explanatory power remains low across all specifications.

Random Forest Model: Predictive Benchmark

As a nonlinear complement to OLS, a Random Forest regressor (500 trees, `max_depth`$= 10$, time-series cross-validation) is trained to predict ΔVIX using rhetoric features (Rhetoric Index, VADER mean, BERT mean, uncertainty ratio, optimistic ratio), tweet volume, Fed communication dummy, and lagged VIX. Despite its flexibility, the RF does not substantially outperform the naive benchmark of predicting $\Delta\text{VIX} = 0$. This confirms that the predictive content of daily rhetoric signals is weak regardless of the modeling approach, and that the variation in ΔVIX is largely driven by market microstructure dynamics rather than presidential communication.



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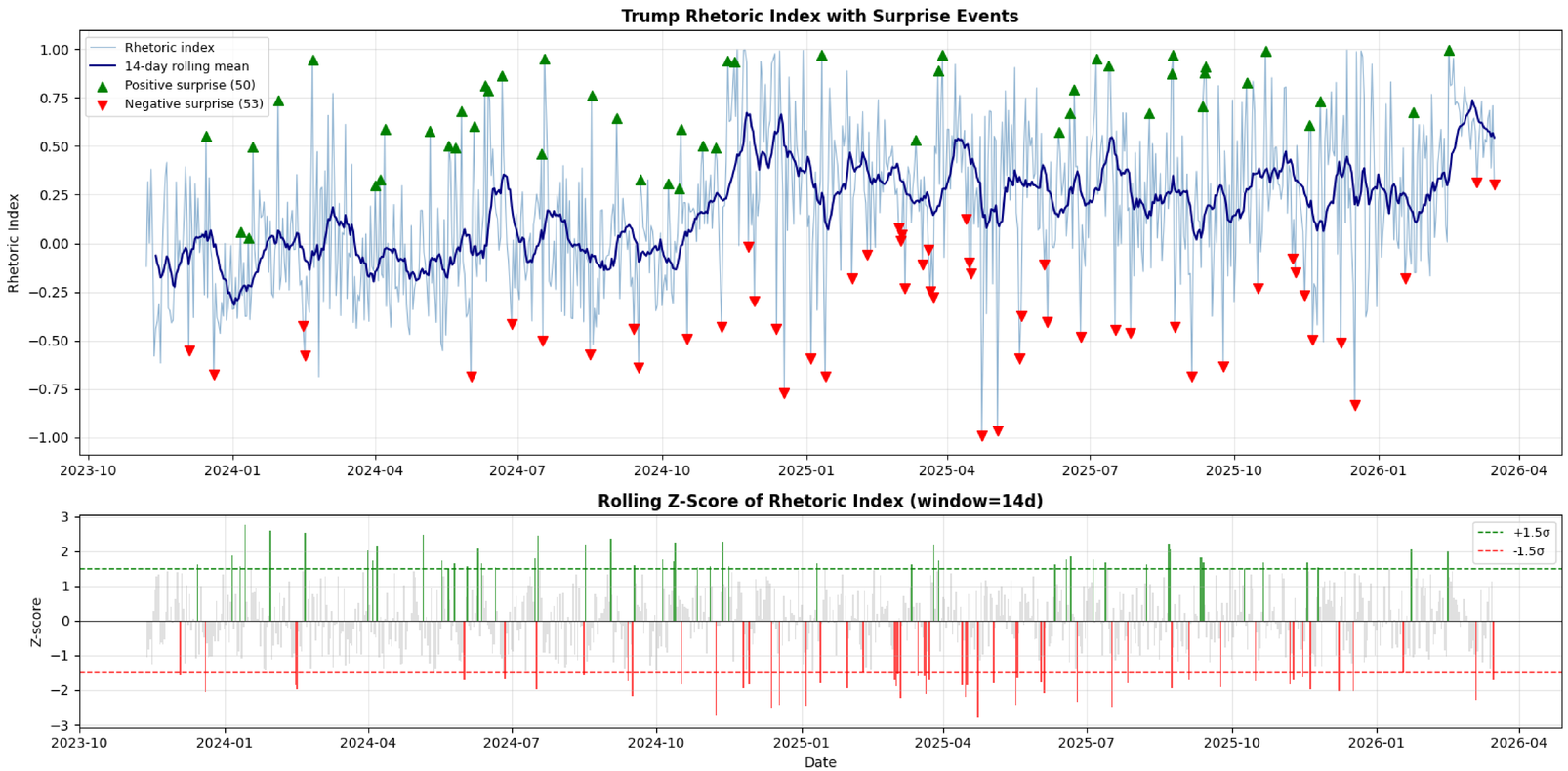
Event Study — Rhetoric Shocks and VIX Changes

To identify *information shocks*, we compute rolling Z-scores of the Rhetoric Index over a 14-day window and classify days with $|z| > 1.5$ as surprise events — either unexpectedly optimistic or unexpectedly aggressive. Cumulative Abnormal VIX changes (CAR) are computed over a $[-2, +5]$ trading-day window, where the abnormal change is defined as the residual from an OLS normal-return model estimated on non-event days. The figure below shows CAR paths with 95% confidence intervals for positive (green) and negative (red) surprise events.

Confidence intervals span zero at all horizons for both event types, and no Average Abnormal Return (AAR) reaches statistical significance at the 5% level. Robustness checks across alternative Z-score thresholds ($|z| \in \{1.0, 1.25, 1.5, 1.75, 2.0\}$) and rolling windows (3–9 days). The event study and OLS results together point to the same conclusion: presidential tweets do not generate detectable next-day shifts in implied volatility at the daily frequency. This finding is consistent with the literature: Machus et al. (2022) find that Trump's tweets cause increased trading activity but *no lasting effects* on prices; Tillmann (2023) shows that VIX effects peak within **two hours** of a tweet and dissipate completely by end of day - rendering daily event windows uninformative by construction.



The figure above plots the smoothed Presidential Rhetoric Index alongside the VIX over the full sample. Periods of elevated risk rhetoric coincide visually with spikes in implied volatility around major political events. However, this visual correlation does not translate into statistically significant linear predictability at the daily close-to-close frequency, as confirmed by the OLS results.
The daily Rhetoric Index has a mean of 0.174 (std. 0.384), ranging from −0.989 to 0.997, indicating substantial variation in presidential tone. Average tweet volume is 12.2 tweets/day (max 77), and the VIX averages 17.1 over the sample, with a maximum of 52.3 during peak uncertainty episodes.

Limitations

- **Data granularity.** Daily VIX data cannot capture intra-day reactions. Tillmann (2023) and Filippou & Gozluklu (2025) show tweet effects peak within 1–2 hours and revert before market close. Intra-day or options microstructure data would be needed to detect short-lived shocks.
- **Inherent unpredictability of VIX.** Forecasting ΔVIX is among the hardest tasks in financial econometrics (Koopman et al., 2005). Market-wide risk appetite, options positioning, and macro surprises dominate any single communication channel.
- **Sentiment model limitations.** VADER and Zero-Shot BERT were not fine-tuned on financial or political text. Trump's rhetoric is highly idiosyncratic — hyperbole, sarcasm, and coded language may be systematically misclassified. A tweet labeled "Optimistic" by the model may be perceived as threatening by markets, and vice versa.
- **Aggregation loss.** Averaging sentiment across all daily tweets collapses within-day heterogeneity. A single high-impact tweet is diluted by dozens of routine posts, attenuating the true signal. Machus et al. (2022) show that tweet-level intra-minute data are required to identify causal effects.
- **Market anticipation.** Markets may price in Trump's communication in real time via news wires and algorithmic monitoring, leaving no predictable residual for next-day models (Born et al., 2017).
- **Potential endogeneity** presidential tweets and market volatility may react to common underlying news or shocks, complicating causal interpretation.
- **Single platform and president.** Results may not generalize to other leaders, communication styles, or platforms. Text-based index may miss non-verbal or institutional communication channels (press conferences, official statements).

Conclusion

Across OLS regressions, robustness checks with decomposed sentiment, and a rolling Z-score event study, Trump's social media rhetoric does not significantly predict next-day VIX changes. The only robust predictor is lagged volatility, consistent with mean-reversion in implied volatility. This result is consistent with recent evidence that tweet effects on volatility are *transitory and intra-day* (Tillmann, 2023; Machus et al., 2022): by the time VIX is recorded at market close, the information contained in tweets has already been fully absorbed. Future work should exploit intra-day VIX data, fine-tune NLP models on financial-political text, and control for concurrent macro announcements to recover the short-lived impact of presidential communication on market uncertainty.

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